K.R. MANGALAM UNIVERSITY

THE COMPLETE WORLD OF EDUCATION

ADA PROJECT

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Date: 03/02/26

Course Code – ENCS256

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Question 1: Fibonacci's Series

```java
import java.util.Scanner;
public class fibonacci {
    static int recursionSteps = 0;  3 usages
    public static long fibonacciRecursive(int n) {  3 usages
        recursionSteps++;
        if (n == 0) return 0; if (n == 1)  return 1;
        return fibonacciRecursive( n: n - 1) + fibonacciRecursive( n: n - 2);
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter number of terms: ");
        int n = sc.nextInt();
        System.out.println("\n--- Fibonacci Series (Iterative) ---");
        long a = 0, b = 1;
        int iterationSteps = 0;
        for (int i = 0; i < n; i++) {
            System.out.print(a + " ");
            long next = a + b;
            a = b;
            b = next;
            iterationSteps++;
        }
        System.out.println("\n\n--- Fibonacci Series (Recursive) ---");
        recursionSteps = 0;
        for (int i = 0; i < n; i++) {System.out.print(fibonacciRecursive(i) + " ");
        }
        System.out.println("\n\nIteration steps: " + iterationSteps); System.out.println("Total recursive calls: " + recursionSteps);
        sc.close();
    }
}
```

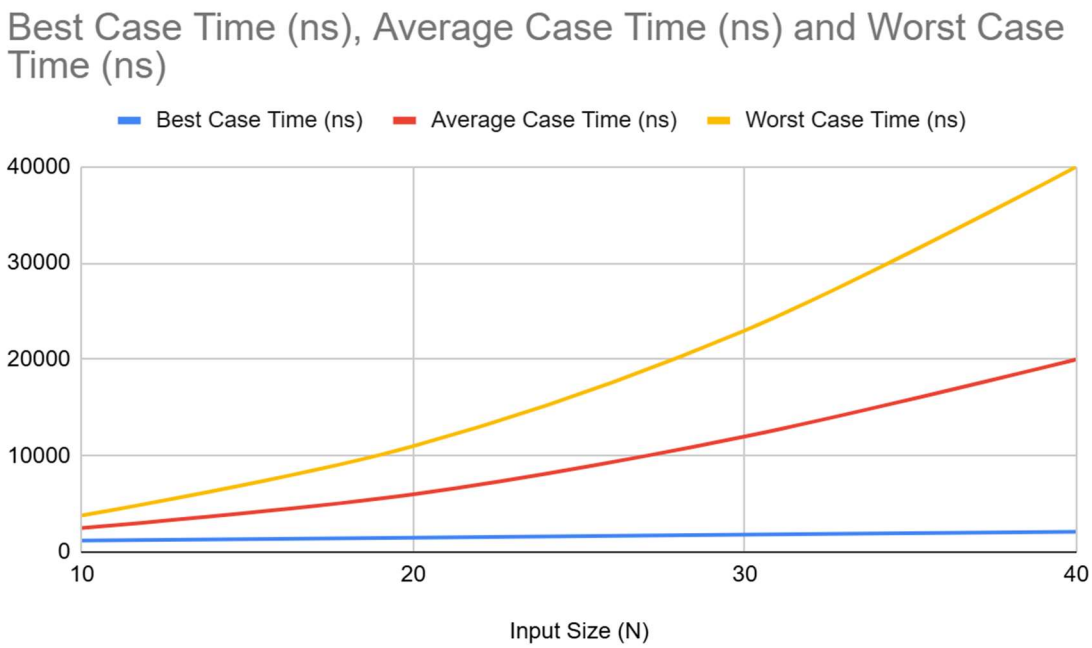
```
C:\Users\ABHISHEK\.jdks\openjdk-25.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=60729" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encod
Enter number of terms: 3

--- Fibonacci Series (Iterative) ---
0 1 1

--- Fibonacci Series (Recursive) ---
0 1 1

Iteration steps: 3
Total recursive calls: 5
```

Question 2: Factorial Number:

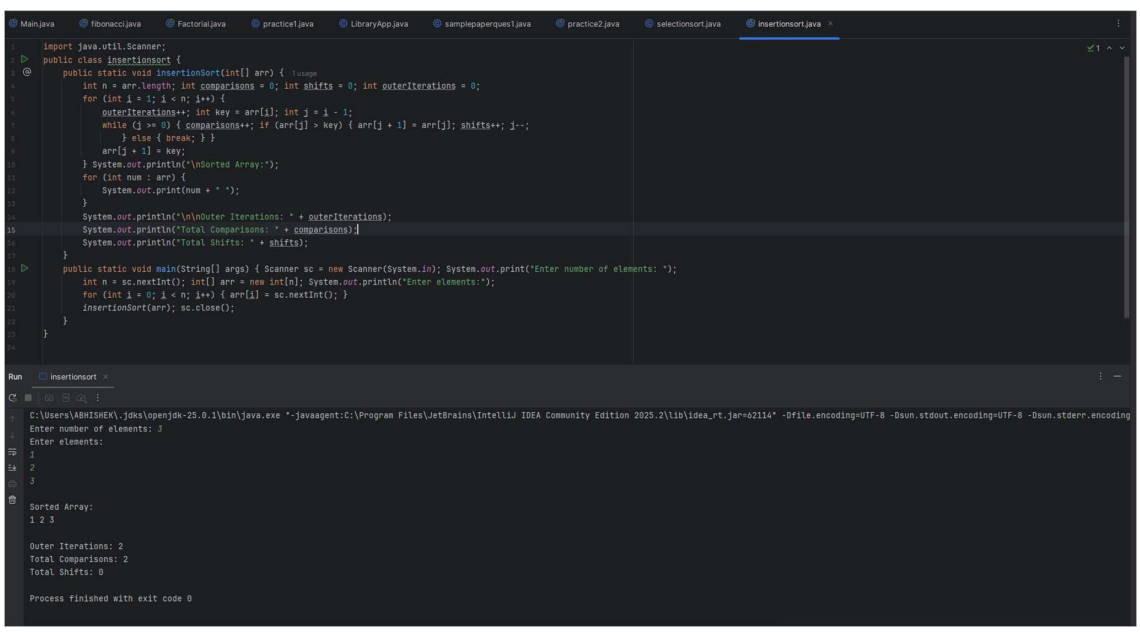
```java
public class Factorial {
    static int stepCount = 0;  6 usages
    static void factIterative(int n) {  1 usage
        stepCount = 0;
        int factorial = 1;
        for (int i = 1; i <= n; i++) {
            factorial = factorial * i;
            stepCount++;
        }
        System.out.println("Iterative Factorial: " + factorial);
        System.out.println("Step Count: " + stepCount);
    }
    static int factRecursive(int n) {  2 usages
        stepCount++;
        if (n <= 1)
            return 1;
        return n * factRecursive( n: n - 1);
    }
    public static void main(String[] args) {
        int n = 5;
        factIterative(n);
        stepCount = 0;
        int result = factRecursive(n);
        System.out.println("Recursive Factorial: " + result);
        System.out.println("Step Count: " + stepCount);
    }
}
```

```
C:\Users\ABHISHEK\.jdks\openjdk-25.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=54451" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding
Iterative Factorial: 120
Step Count: 5
Recursive Factorial: 120
Step Count: 5

Process finished with exit code 0
```

Graph:



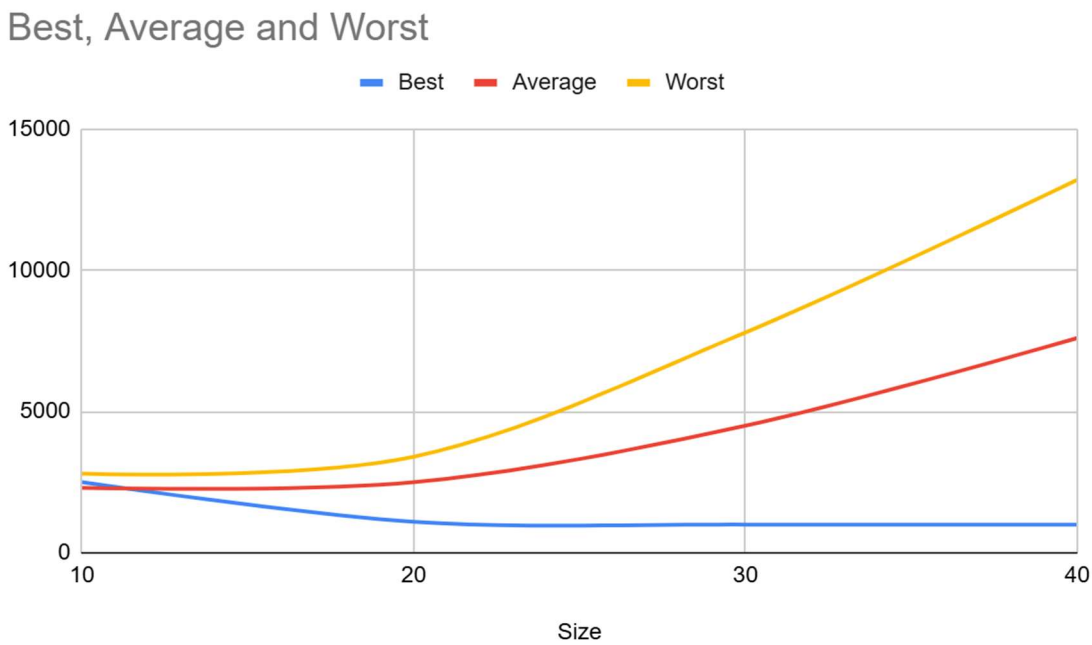
Question 3: Insertion Sort



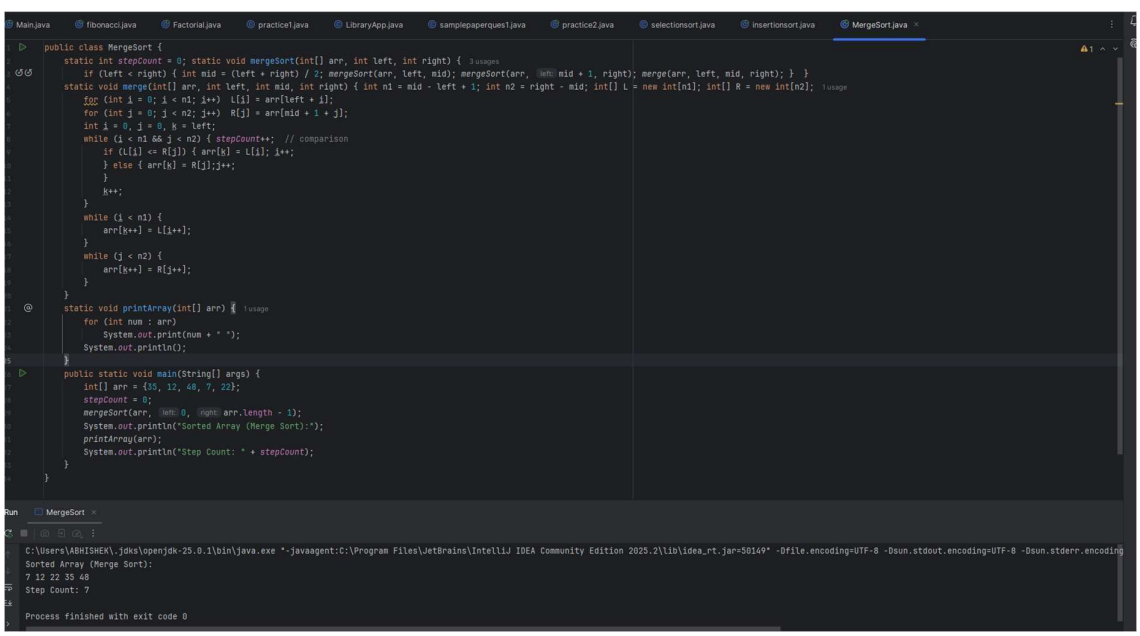
```java
import java.util.Scanner;
public class insertionsort {
    public static void insertionSort(int[] arr) {
        int n = arr.length; int comparisons = 0; int shifts = 0; int outerIterations = 0;
        for (int i = 1; i < n; i++) {
            outerIterations++; int key = arr[i]; int j = i - 1;
            while (j >= 0) { comparisons++; if (arr[j] > key) { arr[j + 1] = arr[j]; shifts++; j--;
                } else { break; } }
            arr[j + 1] = key;
        } System.out.println("\nSorted Array:");
        for (int num : arr) {
            System.out.print(num + " ");
        }
        System.out.println("\n\nOuter Iterations: " + outerIterations);
        System.out.println("Total Comparisons: " + comparisons);
        System.out.println("Total Shifts: " + shifts);
    }
    public static void main(String[] args) { Scanner sc = new Scanner(System.in); System.out.print("Enter number of elements: ");
        int n = sc.nextInt(); int[] arr = new int[n]; System.out.println("Enter elements:");
        for (int i = 0; i < n; i++) { arr[i] = sc.nextInt(); }
        insertionSort(arr); sc.close();
    }
}
```

```
C:\Users\ABHISHEK\.jdks\openjdk-25.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=62114" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding
Enter number of elements: 3
Enter elements:
1
2
3

Sorted Array:
1 2 3

Outer Iterations: 2
Total Comparisons: 2
Total Shifts: 0

Process finished with exit code 0
```

Graph :

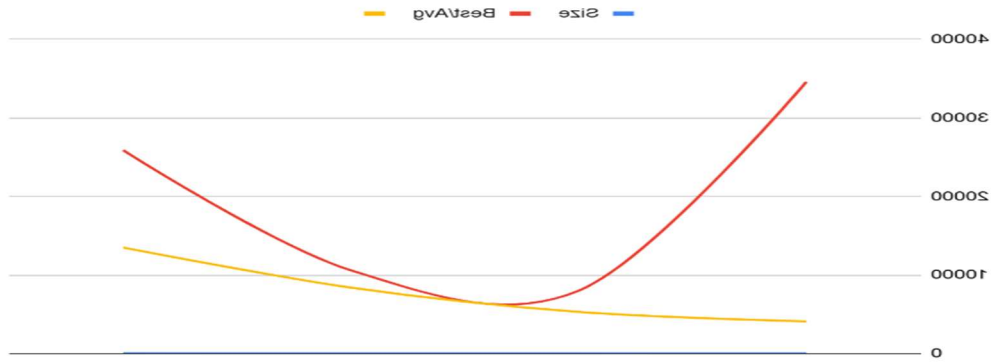


Question 4: Merge Sort

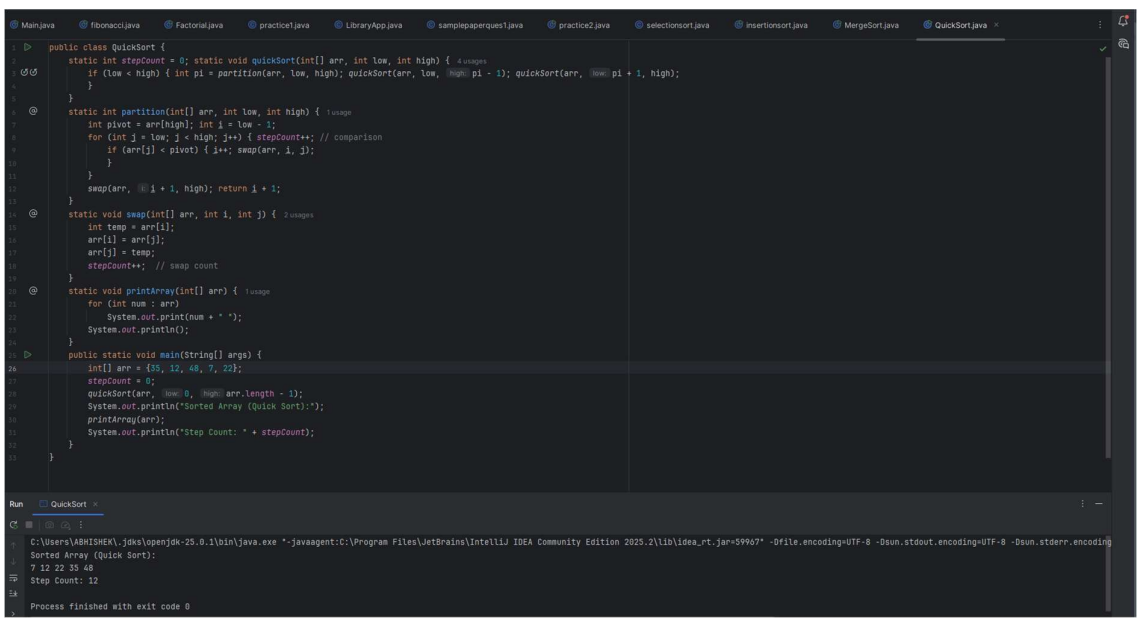


```
Main.java  fibonacci.java  Factorial.java  practice1.java  LibraryApp.java  samplepaperques1.java  practice2.java  selectionsort.java  insertionsort.java  MergeSort.java

public class MergeSort {
    static int stepCount = 0; static void mergeSort(int[] arr, int left, int right) {  3 usages
        if (left < right) { int mid = (left + right) / 2; mergeSort(arr, left, mid); mergeSort(arr, left: mid + 1, right); merge(arr, left, mid, right); }  }
    static void merge(int[] arr, int left, int mid, int right) { int n1 = mid - left + 1; int n2 = right - mid; int[] L = new int[n1]; int[] R = new int[n2];  1 usage
        for (int i = 0; i < n1; i++)  L[i] = arr[left + i];
        for (int j = 0; j < n2; j++)  R[j] = arr[mid + 1 + j];
        int i = 0, j = 0, k = left;
        while (i < n1 && j < n2) { stepCount++;  // comparison
            if (L[i] <= R[j]) { arr[k] = L[i]; i++;
            } else { arr[k] = R[j];j++;
            }
            k++;
        }
        while (i < n1) {
            arr[k++] = L[i++];
        }
        while (j < n2) {
            arr[k++] = R[j++];
        }
    }
    static void printArray(int[] arr) {  1 usage
        for (int num : arr)
            System.out.print(num + " ");
        System.out.println();
    }
    public static void main(String[] args) {
        int[] arr = {35, 12, 48, 7, 22};
        stepCount = 0;
        mergeSort(arr, left: 0, right: arr.length - 1);
        System.out.println("Sorted Array (Merge Sort):");
        printArray(arr);
        System.out.println("Step Count: " + stepCount);
    }
}

Run  MergeSort

C:\Users\ABHISHEK\.jdks\openjdk-25.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=50149" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding
Sorted Array (Merge Sort):
7 12 22 35 48
Step Count: 7

Process finished with exit code 0
```

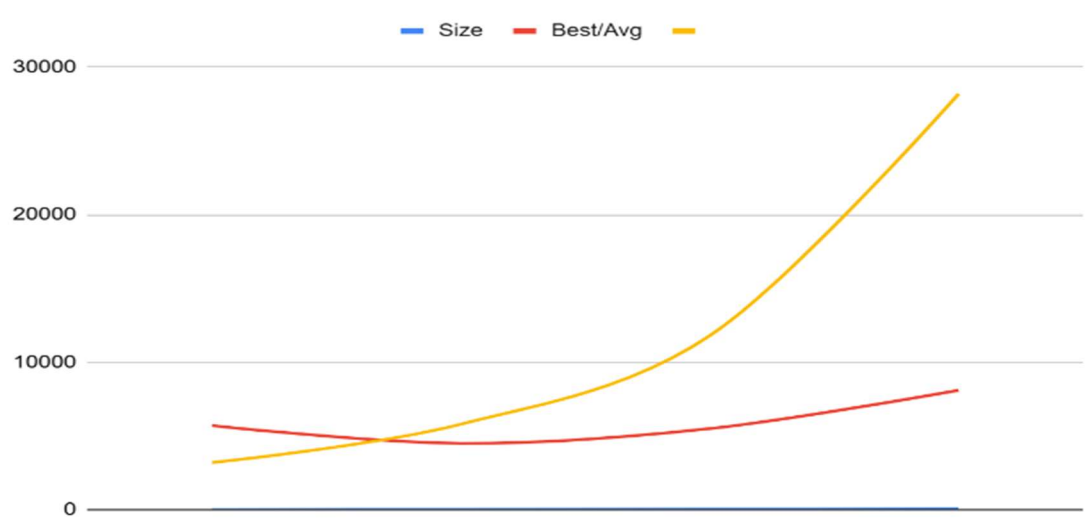
Graph:



Question 5: Quick Sort



Graph:

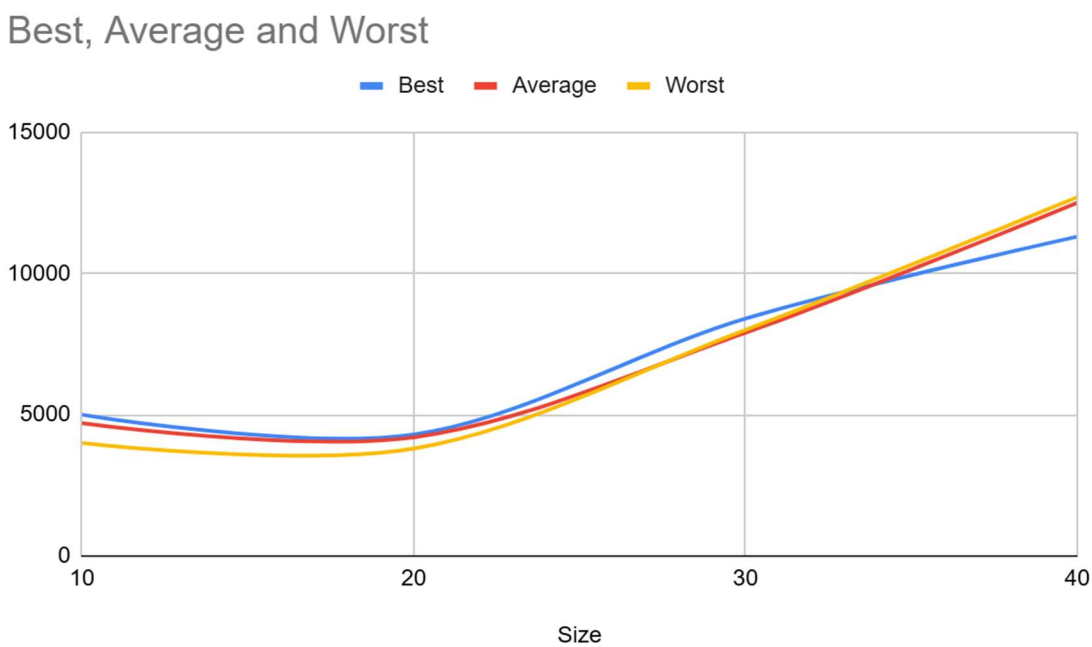


Question 6: Selection Sort

```java
import java.util.Scanner;
public class selectionsort {
    public static void selectionSort(int[] arr) {
        int n = arr.length; int comparisonSteps = 0; int swapSteps = 0;
        for (int i = 0; i < n - 1; i++) { int minIndex = i;
            for (int j = i + 1; j < n; j++) { comparisonSteps++;
                if (arr[j] < arr[minIndex]) { minIndex = j;
                }
            }
            if (minIndex != i) { int temp = arr[i]; arr[i] = arr[minIndex]; arr[minIndex] = temp; swapSteps++;
            }
        }
        System.out.println("\nSorted Array:");
        for (int num : arr) {
            System.out.print(num + " ");
        }
        System.out.println("\n\nTotal Comparisons: " + comparisonSteps); System.out.println("Total Swaps: " + swapSteps);
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in); System.out.print("Enter number of elements: ");
        int n = sc.nextInt(); int[] arr = new int[n]; System.out.println("Enter elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        selectionSort(arr); sc.close();
    }
}
```

```
C:\Users\ABHISHEK\.jdks\openjdk-25.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=49428" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding
Enter number of elements: 3
Enter elements:
12
13
14

Sorted Array:
12 13 14

Total Comparisons: 3
Total Swaps: 0

Process finished with exit code 0
```

Graph:



Question 7: Bubble Sort

```java
public class BubbleSort {
    public static void main(String[] args) {
        int[] arr = {9, 5, 1, 4, 3};
        bubbleSort(arr);
        for (int num : arr) {
            System.out.print(num + " ");
        }
    }
    public static void bubbleSort(int[] arr) {
        int n = arr.length;
        boolean swapped;
        for (int i = 0; i < n - 1; i++) {
            swapped = false;
            for (int j = 0; j < n - i - 1; j++) {
                if (arr[j] > arr[j + 1]) {
                    int temp = arr[j];
                    arr[j] = arr[j + 1];
                    arr[j + 1] = temp;
                    swapped = true;
                }
            }
            if (!swapped) break;
        }
    }
}
```

```
C:\Users\ABHISHEK\.jdks\openjdk-25.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=57423" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding
1 3 4 5 9
Process finished with exit code 0
```

Graph:

Question 8: Linear Search

```java
import java.util.concurrent.ThreadLocalRandom;
import java.util.Scanner;
public class LinearSearch {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int[] arr = new int[5];
        System.out.println("Enter array element: ");
        for(int i=0; i<arr.length; i++) {
            arr[i] = sc.nextInt();
        }
        System.out.println("\nArray elements");
        for(int i=0; i<arr.length; i++) {
            System.out.print(i + " ");
        }
        System.out.println("\nLinear Search");
        linearSearch(arr, target: 20);
    }
    public static void linearSearch(int[] arr, int target) {
        boolean isFound = false;
        for(int i=0; i<arr.length; i++) {
            if(arr[i] == target) {
                System.out.println("element found on: " + arr[i]);
                isFound = true;
                break;
            }
        }
        if(!isFound) System.out.println("element not found in list!");
    }
}
```

```
C:\Users\ABHISHEK\.jdks\openjdk-25.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=65256" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding
Enter array element:
2
3
4

Array elements
0 1 2
Linear Search
element not found in list!

Process finished with exit code 0
```

Question 9: Binary Search

```java
public class BinarySearch {
    static int stepCount = 0;
    static int binarySearchIterative(int[] arr, int key) { stepCount = 0; int low = 0; int high = arr.length - 1;
        while (low <= high) { stepCount++; int mid = (low + high) / 2;
            if (arr[mid] == key)  return mid;
            else if (arr[mid] < key)  low = mid + 1;
            else high = mid - 1;
        } return -1;
    }
    static int binarySearchRecursive(int[] arr, int low, int high, int key) { stepCount++;
        if (low > high) return -1; int mid = (low + high) / 2;
        if (arr[mid] == key) return mid;
        else if (arr[mid] < key)  return binarySearchRecursive(arr, low: mid + 1, high, key);
        else return binarySearchRecursive(arr, low, high: mid - 1, key);
    }
    public static void main(String[] args) {int[] arr = {7, 12, 22, 35, 48}; int key = 22;int result1 = binarySearchIterative(arr, key);
        if (result1 != -1) System.out.println("Iterative: Element found at index " + result1);
        else
            System.out.println("Iterative: Element not found");
        System.out.println("Step Count (Iterative): " + stepCount);
        stepCount = 0;
        int result2 = binarySearchRecursive(arr, low: 0, high: arr.length - 1, key);
        if (result2 != -1)
            System.out.println("Recursive: Element found at index " + result2);
        else
            System.out.println("Recursive: Element not found"); System.out.println("Step Count (Recursive): " + stepCount);
    }
}
```

```
C:\Users\ABHISHEK\.jdks\openjdk-25.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=58602" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding
Iterative: Element found at index 2
Step Count (Iterative): 1
Recursive: Element found at index 2
Step Count (Recursive): 1

Process finished with exit code 0
```

Graph:

Question 10 : Bellman-Ford

```java
import java.util.Scanner;
class Edge {  int u, v, w; Edge(int u, int v, int w) { this.u = u; this.v = v; this.w = w; } }
public class Bellman_Ford { static void bellmanFord(Edge[] edges, int V, int E, int src) { int[] dist = new int[V];
        for (int i = 0; i < V; i++) { dist[i] = Integer.MAX_VALUE;
        } dist[src] = 0;
        for (int i = 1; i <= V - 1; i++) { for (int j = 0; j < E; j++) { int u = edges[j].u; int v = edges[j].v; int w = edges[j].w;
                if (dist[u] != Integer.MAX_VALUE && dist[u] + w < dist[v]) { dist[v] = dist[u] + w;} } }
        for (int j = 0; j < E; j++) {int u = edges[j].u;int v = edges[j].v;int w = edges[j].w;
            if (dist[u] != Integer.MAX_VALUE && dist[u] + w < dist[v]) { System.out.println("Negative weight cycle detected!");return;}}
        System.out.println("Vertex\tDistance"); for (int i = 0; i < V; i++) {System.out.println(i + "\t" + dist[i]);} }
    public static void main(String[] args) { Scanner sc = new Scanner(System.in); int V, E;
        do { System.out.print("Enter number of vertices (<4): ");V = sc.nextInt();
        } while (V <= 0 || V >= 4); do {System.out.print("Enter number of edges (<4): ");E = sc.nextInt();
        } while (E <= 0 || E >= 4);Edge[] edges = new Edge[E];System.out.println("Enter edges (u v w):");
        for (int i = 0; i < E; i++) {int u = sc.nextInt();int v = sc.nextInt();int w = sc.nextInt();
            if (u >= V || v >= V || u < 0 || v < 0) {System.out.println("Invalid vertex! Re-enter edge:");i--;continue;
            }edges[i] = new Edge(u, v, w);}
        int src; do {
            System.out.print("Enter source vertex: ");
            src = sc.nextInt();
        } while (src < 0 || src >= V);
        bellmanFord(edges, V, E, src);
        sc.close();
    }
}
```

```
C:\Users\ABHISHEK\.jdks\openjdk-25.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=52554" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding
Enter number of vertices (<4): 3
Enter number of edges (<4): 3
Enter edges (u v w):
0 1 4
1 2 -2
0 2 5
Enter source vertex: 0
Vertex  Distance
0   0
1   4
2   2

Process finished with exit code 0
```

Question 11: Dijkstra Algorithm

```java
import java.util.Scanner;
public class DijkstraAlgorithm {
    static int minDistance(int[] dist, boolean[] visited, int V) { int min = Integer.MAX_VALUE, min_index = -1;
        for (int v = 0; v < V; v++) { if (!visited[v] && dist[v] <= min) { min = dist[v]; min_index = v; }}
        return min_index;
    }
    static void dijkstra(int[][] graph, int V, int src) { int[] dist = new int[V];boolean[] visited = new boolean[V];
        for (int i = 0; i < V; i++) {dist[i] = Integer.MAX_VALUE;visited[i] = false;
        }dist[src] = 0;
        for (int count = 0; count < V - 1; count++) {int u = minDistance(dist, visited, V);if (u == -1) break;visited[u] = true;
            for (int v = 0; v < V; v++) {if (!visited[v] && graph[u][v] != 0 && dist[u] != Integer.MAX_VALUE && dist[u] + graph[u][v] < dist[v]) {dist[v] = dist[u] + graph[u][v];}}}
        System.out.println("Vertex\tDistance");
        for (int i = 0; i < V; i++) {System.out.println(i + "\t" + dist[i]);}}
    public static void main(String[] args) {Scanner sc = new Scanner(System.in);int V;
        do {System.out.print("Enter number of vertices (<4): ");V = sc.nextInt();
        } while (V <= 0 || V >= 4); int[][] graph = new int[V][V];System.out.println("Enter adjacency matrix:");
        for (int i = 0; i < V; i++) {for (int j = 0; j < V; j++) {graph[i][j] = sc.nextInt();}}int src;
        do { System.out.print("Enter source vertex: ");src = sc.nextInt();
        } while (src < 0 || src >= V); dijkstra(graph, V, src);
        sc.close();
    }
}
```

```
C:\Users\ABHISHEK\.jdks\openjdk-25.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=59256" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding
Enter number of vertices (<4): 3
Enter adjacency matrix:
0 1 4
1 0 2
4 2 0
Enter source vertex: 0
Vertex  Distance
0   0
1   1
2   3

Process finished with exit code 0
```

Question 12: Fractional Knapsack

```java
import java.util.*;
class Item { int value, weight; Item(int value, int weight) {this.value = value; this.weight = weight;} }
public class FractionalKnapSack { static double fractionalKnapsack(int W, Item[] arr, int n) { Arrays.sort(arr, (Item a, Item b) -> { double r1 = (double) a.value / a.weight; double r2 = (double) b.value / b.weight;
            return Double.compare(r2, r1);
        }); double totalValue = 0.0;
        for (int i = 0; i < n; i++) { if (W >= arr[i].weight) { W -= arr[i].weight; totalValue += arr[i].value;
            } else { totalValue += arr[i].value * ((double) W / arr[i].weight); break; } }
        return totalValue;
    }
    public static void main(String[] args) { Scanner sc = new Scanner(System.in); int n, W;
        do { System.out.print("Enter number of items (<4): "); n = sc.nextInt();
        } while (n <= 0 || n >= 4); Item[] arr = new Item[n]; System.out.println("Enter value and weight of each item:");
        for (int i = 0; i < n; i++) { int value = sc.nextInt(); int weight = sc.nextInt();
            if (weight <= 0) { System.out.println("Invalid weight! Re-enter:");i--; continue;
            }
            arr[i] = new Item(value, weight);
        }
        System.out.print("Enter capacity of knapsack: ");
        W = sc.nextInt();
        double result = fractionalKnapsack(W, arr, n);
        System.out.println("Maximum value = " + result);
        sc.close();
    }
}
```

```
C:\Users\ABHISHEK\.jdks\openjdk-25.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=59438" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding
Enter number of items (<4): 3
Enter value and weight of each item:
60 10
100 20
120 30
Enter capacity of knapsack: 50
\Maximum value = 240.0

Process finished with exit code 0
```

Question 13: Matrix Chain Multiplication

```java
import java.util.Scanner;
public class MatrixChainMultiplication {
    static int matrixChainMultiplication(int[] p, int n) { int[][] dp = new int[100][100];
        for (int i = 1; i < n; i++) { dp[i][i] = 0;
        }
        for (int L = 2; L < n; L++) { for (int i = 1; i < n - L + 1; i++) { int j = i + L - 1; dp[i][j] = Integer.MAX_VALUE;
                for (int k = i; k < j; k++) { int cost = dp[i][k] + dp[k + 1][j] + p[i - 1] * p[k] * p[j]; if (cost < dp[i][j]) { dp[i][j] = cost; } } } } return dp[1][n - 1];
    }
    public static void main(String[] args) { Scanner sc = new Scanner(System.in); int n;
        do { System.out.print("Enter number of matrices (<4): "); n = sc.nextInt();
        } while (n <= 0 || n >= 4);
        int[] p = new int[n + 1];
        System.out.println("Enter dimensions array (size " + (n + 1) + "):");
        for (int i = 0; i <= n; i++) {
            p[i] = sc.nextInt();
        }
        int result = matrixChainMultiplication(p, n - n + 1);
        System.out.println("Minimum multiplications = " + result);
        sc.close();
    }
}
```

```
C:\Users\ABHISHEK\.jdks\openjdk-25.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=61139" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding
Enter number of matrices (<4): 3
Enter dimensions array (size 4):
10 20 30 40
Minimum multiplications = 18000

Process finished with exit code 0
```

Question 14: Largest Common Subsequence

```java
import java.util.Scanner;
public class LCS {
    static int lcs(String X, String Y, int m, int n) { int[][] dp = new int[100][100]; for (int i = 0; i <= m; i++) { for (int j = 0; j <= n; j++) { if (i == 0 || j == 0)  dp[i][j] = 0;} } 1 usage
        for (int i = 1; i <= m; i++) {
            for (int j = 1; j <= n; j++) {
                if (X.charAt(i - 1) == Y.charAt(j - 1))
                    dp[i][j] = 1 + dp[i - 1][j - 1];
                else
                    dp[i][j] = Math.max(dp[i - 1][j], dp[i][j - 1]);
            }
        }
        return dp[m][n];
    }
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        String X, Y;
        do {
            System.out.print("Enter first string (length < 4): ");
            X = sc.next();
        } while (X.length() >= 4);
        do {
            System.out.print("Enter second string (length < 4): ");
            Y = sc.next();
        } while (Y.length() >= 4);
        int m = X.length();
        int n = Y.length();
        System.out.println("Length of LCS = " + lcs(X, Y, m, n));
        sc.close();
    }
}
```

```
C:\Users\ABHISHEK\.jdks\openjdk-25.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=52365" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding
Enter first string (length < 4): ab
Enter second string (length < 4): ac
Length of LCS = 1

Process finished with exit code 0
```

Question 15: 0/1 Fractional Knapsack

```java
import java.util.Scanner;
public class ZeroesByOneKnapsack {
    static int knapsack(int W, int[] wt, int[] val, int n) { int[][] dp = new int[100][100];  1 usage
        for (int i = 0; i <= n; i++) {
            for (int w = 0; w <= W; w++) { if (i == 0 || w == 0)  dp[i][w] = 0;
                else if (wt[i - 1] <= w)  dp[i][w] = Math.max( val[i - 1] + dp[i - 1][w - wt[i - 1]], dp[i - 1][w] );
                else  dp[i][w] = dp[i - 1][w]; } } return dp[n][W];}
    public static void main(String[] args) { Scanner sc = new Scanner(System.in); int n, W;
        do { System.out.print("Enter number of items (<4): "); n = sc.nextInt();
        } while (n <= 0 || n >= 4); int[] val = new int[n]; int[] wt = new int[n];
        System.out.println("Enter values of items:");
        for (int i = 0; i < n; i++) {
            val[i] = sc.nextInt();
        }
        System.out.println("Enter weights of items:");
        for (int i = 0; i < n; i++) {
            wt[i] = sc.nextInt();
        }
        System.out.print("Enter capacity of knapsack: ");
        W = sc.nextInt();
        int result = knapsack(W, wt, val, n);
        System.out.println("Maximum value = " + result);
        sc.close();
    }
}
```

```
C:\Users\ABHISHEK\.jdks\openjdk-25.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=62155" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding
Enter number of items (<4): 3
Enter values of items:
60 100 120
Enter weights of items:
10 20 30
Enter capacity of knapsack: 50
Maximum value = 220

Process finished with exit code 0
```

Question 16: BFS

```java
import java.util.*;
public class BFS {
    private int V;  2 usages
    private LinkedList<Integer> adj[];  4 usages
    BFS(int v) {  1 usage
        V = v; adj = new LinkedList[v];
        for (int i = 0; i < v; i++) {adj[i] = new LinkedList<Integer>();} }
    void addEdge(int v, int w) { adj[v].add(w);  6 usages
    }
    void traverse(int s) { boolean visited[] = new boolean[V]; LinkedList<Integer> queue = new LinkedList<Integer>(); visited[s] = true;  1 usage
        queue.add(s);
        while (queue.size() != 0) { s = queue.poll(); System.out.print(s + " "); Iterator<Integer> i = adj[s].listIterator();
            while (i.hasNext()) {
                int n = i.next();
                if (!visited[n]) { visited[n] = true;queue.add(n);
                }
            }
        }
    }
    public static void main(String args[]) {
        BFS g = new BFS( v: 4);
        g.addEdge( v: 0, w: 1);
        g.addEdge( v: 0, w: 2);
        g.addEdge( v: 1, w: 2);
        g.addEdge( v: 2, w: 0);
        g.addEdge( v: 2, w: 3);
        g.addEdge( v: 3, w: 3);
        System.out.println("BFS starting from vertex 2:");
        g.traverse( s: 2);
    }
}
```

```
C:\Users\ABHISHEK\.jdks\openjdk-25.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=50022" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding
BFS starting from vertex 2:
2 0 3 1
Process finished with exit code 0
```

Question 17: DFS

```java
import java.util.*;
public class DFS {  private int V; private LinkedList<Integer> adj[];
    DFS(int v) { V = v; adj = new LinkedList[v];  1 usage
        for (int i = 0; i < v; ++i)  adj[i] = new LinkedList();
    }
    void addEdge(int v, int w) { adj[v].add(w);  6 usages
    }
    void DFSUtil(int v, boolean visited[]) { visited[v] = true; System.out.print(v + " "); Iterator<Integer> i = adj[v].listIterator();  2 usages
        while (i.hasNext()) { int n = i.next();
            if (!visited[n])  DFSUtil(n, visited);
        }
    }
    void traverse(int v) { boolean visited[] = new boolean[V];  1 usage
        DFSUtil(v, visited);
    }
    public static void main(String args[]) {
        DFS g = new DFS( v: 4);
        g.addEdge( v: 0,  w: 1);
        g.addEdge( v: 0,  w: 2);
        g.addEdge( v: 1,  w: 2);
        g.addEdge( v: 2,  w: 0);
        g.addEdge( v: 2,  w: 3);
        g.addEdge( v: 3,  w: 3);
        System.out.println("DFS starting from vertex 2:");
        g.traverse( v: 2);
    }
}
```

```
C:\Users\ABHISHEK\.jdks\openjdk-25.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=57035" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding
DFS starting from vertex 2:
2 0 1 3
Process finished with exit code 0
```

Question 18: N Queens

```java
public class NQueens {  final int N = 4;
    void printSolution(int board[][]) {  1 usage
        for (int i = 0; i < N; i++) {
            for (int j = 0; j < N; j++)  System.out.print(" " + (board[i][j] == 1 ? "Q" : ".") + " "); System.out.println();
        }
    }
    boolean isSafe(int board[][], int row, int col) { int i, j;  1 usage
        for (i = 0; i < col; i++)  if (board[row][i] == 1) return false;
        for (i = row, j = col; i >= 0 && j >= 0; i--, j--)  if (board[i][j] == 1) return false;
        for (i = row, j = col; j >= 0 && i < N; i++, j--)  if (board[i][j] == 1) return false; return true;
    }
    boolean solveNQUtil(int board[][], int col) { if (col >= N) return true;  2 usages
        for (int i = 0; i < N; i++) { if (isSafe(board, i, col)) { board[i][col] = 1;
                if (solveNQUtil(board, col: col + 1)) return true; board[i][col] = 0; }
        }
        return false;
    }
    void solve() { int board[][] = new int[N][N];  1 usage
        if (!solveNQUtil(board, col: 0)) { System.out.print("Solution does not exist"); return;
        }
        printSolution(board);
    }
    public static void main(String args[]) {
        NQueens Queen = new NQueens();
        Queen.solve();
    }
}
```

```
C:\Users\ABHISHEK\.jdks\openjdk-25.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=49900" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding
 . . Q .
 Q . . .
 . . . Q
 . Q . .

Process finished with exit code 0
```

Question 19: Sum of Subsets

```java
import java.util.*;
public class SumOfSubsets {
    static void findSubsets(int[] set, int target, int index, List<Integer> current, int currentSum) {  3 usages
        if (currentSum == target) {
            System.out.println(current);
            return;
        }
        if (currentSum > target || index == set.length) {
            return;
        }
        current.add(set[index]);
        findSubsets(set, target, index: index + 1, current, currentSum: currentSum + set[index]);
        current.remove( index: current.size() - 1);
        findSubsets(set, target, index: index + 1, current, currentSum);
    }
    public static void main(String[] args) {
        int[] set = {10, 7, 5, 18, 12, 20, 15};
        int target = 35;
        System.out.println("Subsets with sum " + target + ":");
        findSubsets(set, target, index: 0, new ArrayList<>(), currentSum: 0);
    }
}
```

```
C:\Users\ABHISHEK\.jdks\openjdk-25.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=55823" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding
Subsets with sum 35:
[10, 7, 18]
[10, 5, 20]
[5, 18, 12]
[20, 15]

Process finished with exit code 0
```

Question 20: Naïve String Matching

Main.java | DFS.java | NQueens.java | SumOfSubsets.java | NaiveStringMatching.java | BFS.java | ZeroesByOneKnapsack.java | fibonacci.java | Factorial.java | LCS.java | MatrixChainMultiplication.java | Fra

```java
public class NaiveStringMatching {
    public static void search(String text, String pattern) {
        int n = text.length();
        int m = pattern.length();
        for (int i = 0; i <= n - m; i++) {
            int j;
            for (j = 0; j < m; j++) {
                if (text.charAt(i + j) != pattern.charAt(j)) {
                    break;
                }
            }
            if (j == m) {
                System.out.println("Pattern found at index " + i);
            }
        }
    }
    public static void main(String[] args) {
        String text = "AABAACAADAABAAABAA";
        String pattern = "AABA";
        System.out.println("Text: " + text);
        System.out.println("Pattern: " + pattern);
        search(text, pattern);
    }
}
```

Run | NaiveStringMatching

```
C:\Users\ABHISHEK\.jdks\openjdk-25.0.1\bin\java.exe "-javaagent:C:\Program Files\JetBrains\IntelliJ IDEA Community Edition 2025.2\lib\idea_rt.jar=51819" -Dfile.encoding=UTF-8 -Dsun.stdout.encoding=UTF-8 -Dsun.stderr.encoding
Text: AABAACAADAABAAABAA
Pattern: AABA
Pattern found at index 0
Pattern found at index 9
Pattern found at index 13

Process finished with exit code 0
```